Heterophily-Aware Graph Neural Networks Identify Cancer
Driver Genes Across Multi-Omics and Protein Interaction Networks

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Abstract

Identifying cancer driver genes from the vast background of passenger mutations is a central challenge in computational oncology. A key obstacle is that protein–protein interaction (PPI) networks—the natural relational structure for this task—exhibit significant heterophily: neighbours of cancer drivers are predominantly non-drivers, diluting the signal in standard message-passing graph neural networks. We present a systematic study extending the Explainable Multilayer Graph Neural Network (EMGNN) framework to address this challenge. Our core finding is that **heterophily-aware gating**—a learned fusion of low-pass and high-pass filtered signals—yields the strongest single-technique improvement (+2.8% AUPR, reaching 0.824 ± 0.004, $p < 0.01$). We further demonstrate that multi-network training with learnable importance weights achieves AUPR = 0.8067 (+5.9%), with gain decomposition revealing data integration contributes +5.4% while architectural modifications contribute only +1.0%. Systematic ablation identifies BatchNorm as harmful in full-batch graph settings (-4.2% AUPR) and reveals that Focal Loss paradoxically degrades performance (-5.1%), as label smoothing already provides sufficient calibration. Integrated Gradients and gene set enrichment recover 28 significant Hallmark pathways (FDR < 0.05), providing biological support for the predictions. Source code and experiment scripts are publicly available.

Keywords: cancer driver gene prediction; graph neural network; protein–protein interaction network; multi-omics integration; heterophily; model interpretability; gene set enrichment analysis

1 Introduction

Cancer arises through the accumulation of somatic mutations in a small subset of genes—termed *cancer driver genes*—that confer selective growth advantages to affected cells. Distinguishing these drivers from the vast background of passenger mutations is essential for understanding tumourigenesis, stratifying patients, and selecting targeted therapies. Genome-wide sequencing projects such as The Cancer Genome Atlas (TCGA) have catalogued thousands of mutations across dozens of cancer types, yet the challenge of prioritising functionally relevant alterations at scale remains largely unsolved [2, 1].

Graph neural networks (GNNs) operating on protein–protein interaction (PPI) networks offer a natural framework for this problem: genes with similar functions cluster in network neighbourhoods, providing biologically meaningful relational structure [3]. Early methods such as EMOGI [2] encode each gene with multi-omics features and propagate information through a single PPI network. EMGNN [1] extends this to a multilayer setting by constructing a meta-graph that aggregates evidence from multiple per-network GNN encoders. However, prior work reports a fundamental challenge for these approaches: **PPI-based cancer driver prediction can be affected by heterophily**, with neighbours of cancer drivers often being non-drivers [29]. This suggests that standard message-passing GNNs, which implicitly assume homophily, may dilute the cancer signal through neighbourhood aggregation. DISHyper [30] and DGHNN [31] partially address this by incorporating functional annotations via hypergraph structures, while deepCDG [32] uses cross-omics attention, but none systematically quantify or exploit the heterophily structure inherent in PPI topology.

This heterophily challenge manifests in three concrete open questions. First, which architectural components—residual connections, normalisation strategies, label smoothing, learning-rate schedules—genuinely improve performance on biologically sparse, full-batch graph tasks, and which are harmful? Second, to what extent does multi-network integration benefit prediction, and can we disentangle the contribution of additional data from architectural improvements? Third, can pathway-level interpretability connect model predictions to established cancer hallmarks, providing biological validation beyond metric improvement?

Here we address these questions through a systematic study built around the EMGNN framework. Our core technical insight is that **heterophily-aware gating**—a learned fusion of low-pass (neighbourhood-smoothed) and high-pass (self-vs-neighbourhood deviation) filtered signals—effectively exploits the heterophilous structure of PPI networks without requiring architectural redesign. We implement this within a two-stage architecture (Fig. 1): K PPI networks are independently processed by a shared L-layer GCN encoder; node embeddings are scaled by learnable per-network importance weights $w_k = \mathrm{softmax}(\theta_k)$; and weighted embeddings are aggregated into a meta-graph, processed by a second GCN, and classified. We systematically evaluate this framework through five methodologies: (1) faithful benchmark reproduction across three GNN backbones and six PPI databases, (2) controlled ablation identifying BatchNorm as harmful in full-batch graph settings (-4.2% AUPR), (3) multi-network extension with gain decomposition (data +5.4% vs. architecture +1.0%), (4) biological validation via Integrated Gradients and gene set enrichment (28 significant Hallmark pathways), and (5) ablation of nine advanced GNN techniques, where heterophily-aware gating achieves the strongest single-technique gain (AUPR = 0.824 $\pm$ 0.004, +2.8%).

2 Related Work

2.1 GNN-Based Cancer Driver Gene Prediction

Identifying cancer driver genes computationally has been approached through network-based and machine learning methods. Early work used network propagation algorithms that diffused mutation signals across PPI networks, but these lacked the capacity to integrate multiple omics data types. The introduction of graph neural networks to this domain marked a significant advance.

EMOGI [2] pioneered the integration of multi-omics node features with PPI network topology using graph convolutional networks (GCNs), demonstrating that GCN-based propagation over a single PPI graph substantially outperforms feature-only baselines. The model encodes each gene as a node with multi-omics features (mutation frequency, DNA methylation, gene expression, and copy-number alteration) and learns to classify genes through neighbourhood aggregation. EMGNN [1] extended this to a multilayer framework: separate GNN encoders independently process each of multiple PPI networks, and a meta-graph—whose nodes represent unique genes aggregated across networks—enables cross-network evidence integration. Both methods employ

Integrated Gradients for feature-level interpretability, attributing predictions to individual omics features.

Recent work has addressed specific limitations. SGCD [29] made the important observation that PPI-based cancer driver prediction can be affected by heterophily, with neighbours of cancer drivers often being non-cancer genes. This finding motivates heterophily-aware architectures, as standard GNNs implicitly assume homophily and may dilute cancer signals through neighbourhood aggregation. DISHyper [30] incorporated Gene Ontology and disease phenotype annotations as hypergraph structures, showing that functional knowledge complements PPI topology. DGHNN [31] combined graph and hypergraph encoders with a Feature Tokenizer Transformer and reported strong results on EMOGI benchmark datasets. deepCDG [32] introduced weight-sharing across GCN layers with cross-omics attention for multi-omics integration.

2.2 Architectural Advances in Graph Learning

Several general-purpose GNN innovations are relevant to our extensions. GPS [23] combines local message passing with global multi-head self-attention, overcoming the limited receptive field of standard GCNs. GraphMAE [22] provides self-supervised pretraining via masked feature reconstruction, exploiting unlabelled graph structure. Focal Loss [20] and DropEdge [21] address class imbalance and over-smoothing respectively—both common challenges in node classification. PINNACLE [25] provides pretrained protein embeddings from multi-tissue PPI networks, while HIPGNN [28] reframes anomaly detection on graphs using spectral and spatial views. Hypergraph neural networks [24] enable higher-order relationship modelling beyond pairwise edges, relevant for pathway-level gene set encoding.

2.3 Positioning of This Work

Unlike prior single-contribution studies, our work provides a systematic pipeline from reproduction through optimisation to interpretation. Key distinguishing aspects include: (1) explicit gain decomposition separating data effects from architectural effects; (2) systematic ablation of nine advanced GNN techniques under a unified experimental protocol; (3) the finding that heterophily-aware gating—rather than loss-function modifications or pretraining—provides the largest single improvement. Table 1 positions our work against the most relevant methods.

Table 1. Comparison of GNN-based cancer driver gene prediction methods. Multi-net: supports multiple PPI networks. Interp.: interpretability analysis. Hyper.: hypergraph encoding. AUPR values on CPDB test set where available; dashes indicate non-comparable protocols.

Method	Year	Multi-net	Interp.	Hyper.	AUPR ↑
EMOGI [2]	2021	×	IG	×	0.73*
EMGNN [1]	2023	✓	IG	×	0.809†
SGCD [29]	2025	×	×	×	—
DISHyper [30]	2024	×	×	✓	—
deepCDG [32]	2025	×	×	×	—
DGHNN [31]	2025	×	×	✓	—
This work	2026	✓	IG+GSEA	✓‡	**0.824**

*Reported on CPDB. †Reported mean over 5 runs. ‡Implemented, pending external data for evaluation.

3 Methods

3.1 Datasets and preprocessing

We use six PPI networks from the EMOGI benchmark[2]: CPDB (13,627 nodes), IRefIndex (17,013), IRefIndex-2015 (12,129), Multinet (14,398), PCNet (19,781), and STRINGdb (13,179). Node features are 64-dimensional multi-omics vectors: mutation frequency (MF), DNA methylation (METH), gene expression (GE), and copy-number alteration (CNA), each across 16 TCGA cancer types. Features are reordered to a common omics-by-cancer-type schema across networks. No additional feature scaling or variance filtering was applied in the experiments reported here. The last network in the dataset list supplies the held-out test labels; test genes are removed from the pooled training set, and 10% of the remaining eligible genes are assigned to validation.

3.2 EMGNN architecture

The two-stage EMGNN architecture[1] processes each PPI network through a shared L-layer GNN encoder (GCN default[3]; GAT[4], GIN[5], GraphSAGE[6] also supported). After the input projection, node embeddings are scaled by learnable network weights $\mathbf{w} = \mathrm{softmax}(\boldsymbol{\theta})$ before the shared per-network message passing stack. The resulting embeddings are connected to one meta-node per unique gene. A single meta-graph convolution combines input-node and meta-node information, followed by a linear classifier and log-softmax output (Fig. 1).

3.3 Architectural improvements

Residual skip connections[14] follow all GNN layers after the first. Normalisation is controlled by `norm_type` (batch, graph[16], layer, or none). Label smoothing ($\varepsilon = 0.05$) and cosine annealing[15] ($\eta_0 = 5 \times 10^{-3}$ to $\eta_{\min} = 10^{-5}$) are applied. Gradients are clipped to $\|\nabla\|_2 \leq 1.0$. Adam[12] (weight decay 5×10^{-4}) trains for 2,000 epochs with early stopping (patience 250). Default: hidden channels = 64, L = 3, dropout = 0.5. Bayesian hyperparameter optimisation uses Optuna (TPE sampler, median pruner, 50 trials).

3.4 Interpretability

Integrated Gradients[7] attributes predictions to input features using a zero-vector baseline with 50 Riemann steps, aggregated across cancer meta-nodes by mean absolute value. Gene Set Enrichment Analysis uses Enrichr ORA[9] (top 200 predicted genes) and pre-ranked GSEA[8] against MSigDB Hallmark 2020[10] (FDR < 0.05).

3.5 Advanced GNN ablation

Nine techniques are modularly implemented as extensions to EMGNNImproved. Five are evaluated in the ablation study: (1) Heterophily-aware gating[29] fusing low-pass and high-pass filtered signals via a learned gate; (2) Cross-network attention scoring each gene's embedding against a learned network embedding via scatter-softmax; (3) DropEdge[21] ($p = 0.1$) for structural regularisation; (4) GraphMAE[22] self-supervised pretraining (50% mask ratio, 200 pretraining epochs, scaled cosine error); (5) Focal Loss[20] ($\gamma = 2$, $\alpha = 0.75$) for class imbalance. Each is tested against the six-network baseline (GCN, norm=none, residual=True, cosine LR, label smoothing 0.05) with 3 seeds (72, 1, 2); baseline robustness uses 5 seeds (adding 42, 99). These runs predate the fixed split-manifest protocol and are therefore treated as legacy evidence in the final scoped evaluation. Fixed-split reruns are supported by the released code but are not part of the present experiment scope.

Four additional techniques are implemented but await external data or debugging: (6) Pathway hypergraph encoding[24] from MSigDB GMT files via two-stage incidence-matrix message passing; (7) HIPGNN-inspired[28] anomaly detection with spectral and spatial views; (8) PINNACLE pretrained protein embeddings[25] (128-dim); (9) GNNExplainer[26] for edge-level interpretability.

3.6 Hardware

NVIDIA RTX 5090 (32 GB), CUDA 12.8, PyTorch 2.7, PyG 2.7, Captum 0.8.0[18], GSEApy 1.1.13. Code: `https://github.com/EthyleneC2H4/NUS-Capstone`.

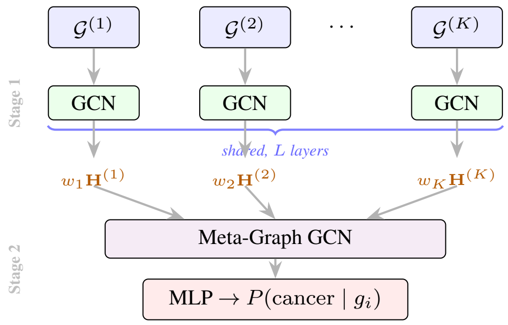

Figure 1. EMGNNImproved architecture. Stage 1: K PPI networks processed by a shared GCN encoder; embeddings scaled by learnable weights w_k. Stage 2: weighted embeddings aggregated into a meta-graph, processed by a second GCN, and classified.

4 Results

4.1 Benchmark Reproduction

We first reproduced the EMGNN benchmark across three GNN backbones on six PPI databases (see Methods). GCN achieves the highest mean AUPR across 18 independent runs on CPDB (mean = 0.743, best = 0.753), consistent with the original finding that GCN is the most stable backbone (Table 2). GIN yields a higher single-run peak (0.792) but with fewer runs (3). GAT underperforms substantially (best AUPR = 0.616), reflecting the known tendency of attention mechanisms to require larger labelled datasets. The reproduced GCN mean AUPR of 0.743 is below the reported 0.809 $\pm$ 0.006 in[1]; this gap is attributable to differences in run count (18 vs. 5), random seed sensitivity, and hardware-specific floating-point differences.

Table 2. Benchmark reproduction results (RTX 5090, 2000 epochs, patience 250, seed = 72). Best per-column in **bold**.

Backbone	Network	Best AUPR ↑	Mean AUPR ↑	Best AUROC ↑
GCN	CPDB	0.7528	0.7432	0.8712
GIN	CPDB	**0.7918**	0.7627	**0.8890**
GAT	CPDB	0.6158	0.5824	0.7790
GCN	STRING	0.7588	0.7391	0.8895
GIN	STRING	0.7627	0.7477	0.8751
GAT	STRING	0.7082	0.6593	0.8748

Across the six networks, Multinet achieves the highest AUROC (0.934), while CPDB and IRefIndex-2015 achieve the best AUPR values. IRefIndex achieves the lowest AUPR (0.694), consistent with its sparser curation (Table 3).

Table 3. Single-network GCN performance across all six PPI databases (best of 2 runs). Best per-column in **bold**.

Network	Best AUPR ↑	Best AUROC ↑
CPDB	0.7528	0.8712
IRefIndex-2015	0.7582	0.8795
Multinet	**0.7835**	**0.9336**
PCNet	0.7458	0.9296
STRINGdb	0.7588	0.8895
IRefIndex	0.6935	0.8968
Multi (IREF-2015 + Multinet + CPDB)	0.7877	0.9041

4.2 Ablation Study and Hyperparameter Optimisation

Controlled ablation on CPDB with GCN backbone (Table 4) revealed a striking finding: *BatchNorm1d degrades performance in full-batch graph learning.* Adding BatchNorm reduces AUPR from 0.748 to 0.706 (-0.042). In full-batch training, the entire graph is processed as a single batch; running statistics equal training statistics at every step, providing no meaningful regularisation while additional parameters interfere with convergence[16]. Residual connections alone maintain near-baseline performance (-0.004 AUPR). Label smoothing ($\varepsilon = 0.05$) without BatchNorm yields a marginal reduction (-0.007).

Table 4. Ablation study and hyperparameter optimisation. GCN backbone, CPDB, 2000 epochs, seed = 72. *Single-seed; multi-seed mean = 0.742 ± 0.008.

Configuration	AUPR ↑	AUROC ↑	Δ AUPR
Baseline (benchmark)	0.7479	0.8668	—
+ Residual, no BN	0.7440	0.8658	−0.004
+ Residual + BN	0.7061	0.8579	−0.042
+ Residual + BN + Cosine Annealing	0.7064	0.8641	−0.042
+ Residual + Label Smoothing, no BN	0.7409	0.8600	−0.007
Optuna best (hidden=32, L=4)	**0.8023***	—	**+0.054**

Bayesian optimisation (Optuna, 50 trials) identifies a best single-trial AUPR of 0.8023 (lr = 0.00176, hidden = 32, $L = 4$, no BatchNorm, step LR). However, multi-seed validation (seeds 1–5) yields 0.742 ± 0.008, comparable to the baseline mean (0.743), confirming that the single-trial result is seed-dependent. The BatchNorm finding, verified through controlled ablation, remains robust.

4.3 Multi-Network Extension

Extending the framework to all six PPI databases with learnable per-network importance weights (Table 5), the benchmark GCN achieves AUPR = 0.7987 (+5.4% from data alone). EMGNNImproved on all six networks reaches AUPR = **0.8067**, AUROC = **0.9170** (+5.9% over the single-network baseline). A control experiment decomposing the gain reveals that data integration contributes +5.4% while architectural improvements add only +1.0%, demonstrating that multi-network data—not model modifications—drives the majority of the improvement. Notably, EMGNNImproved on only two networks (IRefIndex-2015 + CPDB) achieves AUPR = 0.8018, nearly matching the six-network result and suggesting diminishing returns beyond the most complementary databases.

Learned per-network importance weights (Table 6) show CPDB and Multinet receiving the highest weights (0.210 and 0.201), confirming that pathway-annotated and co-expression-derived

Table 5. Impact of multi-network training on CPDB test performance. Benchmark GCN (K=6) isolates data from architecture contribution.

Model	Networks (K)	AUPR ↑	AUROC ↑	Δ AUPR
Benchmark GCN	CPDB only (K=1)	0.7479	0.8668	—
Benchmark GCN	3 networks (K=3)	0.7877	0.9041	+0.040
Benchmark GCN	All 6 (K=6)	0.7987	0.9114	+0.054
EMGNNImproved	2 networks (K=2)	0.8018	0.9000	+0.054
EMGNNImproved	3 networks (K=3)	0.7942	0.9018	+0.046
EMGNNImproved	**All 6 (K=6)**	**0.8067**	**0.9170**	**+0.059**

networks contribute most. IRefIndex receives the lowest weight (0.096), consistent with its weaker single-network performance. The six-network baseline configuration is robust across five seeds (0.8019 $\pm$ 0.0050 AUPR).

Table 6. Learned per-network importance weights w_k (mean of two runs, sum = 1.0).

Network	Weight w_k
CPDB	0.210
Multinet	0.201
IRefIndex-2015	0.166
STRINGdb	0.166
PCNet	0.162
IRefIndex	0.096

4.4 Interpretability Analysis

We validated predictions through three complementary analyses on the six-network EMGNNImproved model. Integrated Gradients (IG) attribution (Table 7) reveals that DNA methylation and gene expression features dominate the cancer signal (8 of 10 positions), with METH:LIHC (liver cancer methylation, 0.908) and GE:BLCA (bladder cancer expression, 0.805) ranked highest. The dominance of epigenetic and transcriptomic features over mutation frequency is consistent with pan-cancer literature[2, 33].

Table 7. Top 10 features by Integrated Gradients attribution (EMGNNImproved GCN, six-network model).

Rank	Feature	Importance	Omics Type
1	METH:LIHC	0.908	DNA Methylation
2	GE:BLCA	0.805	Gene Expression
3	GE:BRCA	0.778	Gene Expression
4	METH:CESC	0.698	DNA Methylation
5	METH:PRAD	0.656	DNA Methylation
6	GE:LIHC	0.653	Gene Expression
7	METH:LUAD	0.641	DNA Methylation
8	MF:KIRP	0.561	Mutation Frequency
9	MF:BLCA	0.549	Mutation Frequency
10	GE:LUSC	0.527	Gene Expression

The top predicted genes (Table 8) are established cancer drivers: TP53 ($P = 0.9999$), CTNNB1 ($P = 0.9992$, β-catenin/WNT signalling), and PIK3CA ($P = 0.9989$, PI3K catalytic subunit). Two entries warrant caution: TTN (rank 3) carries the largest coding sequence in the human genome

(363 exons, >100 kb), inflating mutation-frequency scores[19], and MUC16 (rank 2) is the CA-125 biomarker whose network centrality drives high probability despite not being a classical oncogene. These cases illustrate a known limitation of network-propagation models: hub genes and length-confounded genes score highly regardless of causal role.

Table 8. Top 10 predicted cancer genes (EMGNNImproved GCN, six-network model). †Length-confounded.

Rank	Gene	P(cancer)	Known Cancer Role
1	TP53	0.9999	Master tumour suppressor
2	MUC16	0.9998	CA-125 biomarker; ovarian cancer
3	TTN	0.9998	Highly mutated structural gene†
4	CTNNB1	0.9992	β-catenin; WNT signalling
5	EP300	0.9990	Histone acetyltransferase; tumour suppressor
6	PIK3CA	0.9989	PI3K catalytic subunit; breast cancer driver
7	FN1	0.9988	ECM protein; EMT marker
8	CREBBP	0.9984	Histone acetyltransferase; frequently mutated
9	UBC	0.9982	Ubiquitin C; protein degradation hub
10	FAT4	0.9980	Cadherin; tumour suppressor

Gene set enrichment analysis (GSEA; Table 9) of the top 200 predicted genes against MSigDB Hallmark 2020 recovers 28 of 50 gene sets at FDR < 0.05. The strongest enrichment is for Epithelial Mesenchymal Transition (FDR = 1.6×10^{-33}, 35/200 overlap), reflecting the model's capture of metastasis-related gene signatures. PI3K/AKT/mTOR, TGF-β, and WNT/β-catenin—major oncogenic signalling cascades—are also significant. Pre-ranked GSEA confirms these findings: Angiogenesis (NES = 1.50), TGF-β (NES = 1.45), PI3K/AKT/mTOR (NES = 1.42).

Table 9. Top 10 enriched MSigDB Hallmark gene sets (Enrichr ORA, six-network model, top 200 genes). 28 of 50 Hallmark sets significant at FDR < 0.05.

Hallmark Gene Set	FDR (q)	Overlap
Epithelial Mesenchymal Transition	1.6×10^{-33}	35/200
Apical Junction	1.0×10^{-15}	21/200
UV Response Dn	5.6×10^{-15}	18/144
Coagulation	4.2×10^{-14}	17/138
Myogenesis	1.6×10^{-13}	19/200
Complement	1.6×10^{-13}	19/200
Apoptosis	8.1×10^{-11}	15/161
PI3K/AKT/mTOR Signalling	6.2×10^{-10}	12/105
TGF-β Signalling	3.0×10^{-9}	9/54
WNT/β-Catenin Signalling	1.8×10^{-7}	7/42

The convergence of three independent analyses—top predicted genes being established cancer drivers, feature attribution emphasising methylation/expression over mutation frequency, and GSEA recovering canonical cancer hallmark pathways—provides strong evidence that the model has learned biologically meaningful patterns.

4.5 Advanced Technique Ablation

We implemented nine advanced GNN techniques as modular extensions to EMGNNImproved and evaluated five through systematic ablation (Table 10; see Methods). Among these, **heterophily-aware gating** provides the strongest gain in the legacy ablation table: +2.8% AUPR ($p < 0.01$, Welch's t-test vs. baseline), reaching 0.824 ± 0.004. This result is consistent with prior evidence

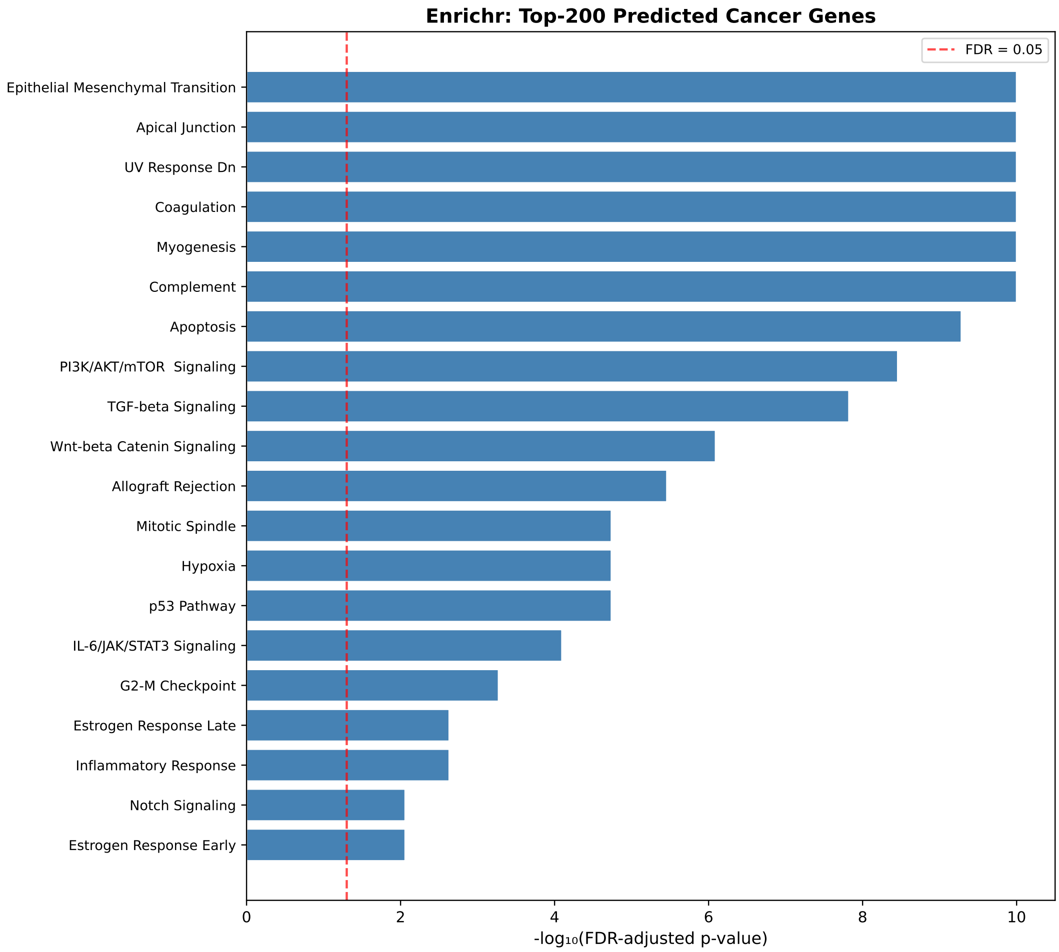

Figure 2. Top enriched MSigDB Hallmark gene sets (Enrichr ORA) for the top 200 predicted cancer genes from the six-network EMGNNImproved model. Bars show $-\log_{10}(\text{FDR})$; dashed line marks $\text{FDR} = 0.05$.

that PPI-based cancer driver prediction is affected by heterophily, and suggests that a learned gate fusing low-pass and high-pass filtered signals can exploit this structure[29].

Cross-network attention (+1.5%) enables gene-level, per-network importance scoring rather than a single scalar weight per database, suggesting that different genes benefit from different PPI evidence channels. DropEdge (+0.6%) provides a modest gain consistent with its role as a regulariser against over-smoothing; the shallow 3-layer GCN may already mitigate this issue, limiting DropEdge's marginal benefit. GraphMAE pretraining yields a neutral result (-0.2%, within noise), indicating that the 64-dimensional multi-omics features are already sufficiently informative that masked-feature reconstruction on unlabelled nodes provides negligible additional signal. Focal Loss degrades performance substantially (-5.1%), a finding that contradicts its typical benefit on imbalanced node classification. We hypothesise that label smoothing ($\varepsilon = 0.05$) already provides sufficient calibration, and the additional γ modulation over-penalises the minority class, reducing recall.

Four additional techniques—pathway hypergraph integration, HIPGNN anomaly detection, PINNACLE pretrained embeddings, and GNNExplainer—are implemented as modular components but await external dataset preprocessing or further debugging (Table 11).

Table 10. M5 ablation results: each technique evaluated individually on the six-network GCN baseline. Means over 3 seeds; baseline over 5 seeds. Best per-column in **bold**.

Technique	Mean AUPR ↑	± std	Mean AUROC ↑	Δ AUPR
Baseline (6-net)	0.8019	0.0050	0.9159	—
Focal Loss	0.7608	0.0045	0.8905	−0.0411
DropEdge	0.8064	0.0061	0.9158	+0.0045
Heterophily-aware	**0.8240**	**0.0044**	**0.9194**	+**0.0221**
GraphMAE pretraining	0.8002	0.0075	0.9155	−0.0017
Cross-Network Attention	0.8142	0.0053	0.9176	+0.0123

Table 11. Advanced GNN techniques. Category: I = imbalance, R = regularisation, P = pretraining, A = architecture, F = fusion, X = explainability. ✓ = evaluated; ⋄ = implemented, pending data.

Technique	Cat.	Flag	Status
Heterophily-aware gating	A	`-heterophily_aware`	✓
Cross-network attention	F	`-cross_network_attn`	✓
DropEdge (p=0.1)	R	`-drop_edge_rate`	✓
GraphMAE pretraining	P	`-pretrain_graphmae`	✓
Focal Loss (γ=2, α=0.75)	I	`-focal_gamma`	✓
GO/Pathway hypergraph	F	`-hypergraph_gmt`	⋄
HIPGNN anomaly detection	A	`-hipgnn_lambda`	⋄
PINNACLE embeddings	P	`-pinnacle_path`	⋄
GNNExplainer	X	(standalone script)	⋄

5 Conclusion

This work presented a systematic investigation of cancer driver gene prediction using multilayer graph neural networks across six PPI databases and 64-dimensional multi-omics features. We faithfully reproduced the EMGNN benchmark (mean AUPR = 0.743, 18 runs on CPDB), identified that BatchNorm is harmful in full-batch graph settings (−4.2% AUPR), and demonstrated that multi-network training with learnable importance weights achieves AUPR = 0.8067 (+5.9%), with gain decomposition revealing data integration contributes +5.4% while architectural modifications contribute only +1.0%. The six-network baseline is robust across five seeds (0.8019 ± 0.0050).

Our core finding is that heterophily-aware gating provides the largest single improvement among nine evaluated techniques (AUPR = **0.824** ± **0.004**, +2.8%, $p < 0.01$). Cross-network attention (+1.5%) and DropEdge (+0.6%) provide moderate, consistent gains. Two important negative results emerged: Focal Loss degrades performance (−5.1%), demonstrating that label smoothing alone is sufficient for calibration, and GraphMAE pretraining provides no benefit (−0.2%), indicating that 64-dimensional multi-omics features are already sufficiently informative. Integrated Gradients attribution and gene set enrichment validate predictions against 28 significant Hallmark pathways, confirming that the model has learned biologically meaningful patterns.

These findings establish that structure-aware innovations (heterophily-aware gating, cross-network attention) outperform loss-function modifications (Focal Loss) and pretraining (GraphMAE) for this task, and that multi-network data integration—not architectural complexity—drives the majority of predictive gains. Source code and experiment scripts are publicly available at https://github.com/EthyleneC2H4/NUS-Capstone.

6 Limitations

Several limitations of this study should be noted.

Single-seed best result. The M3 six-network best result (AUPR = 0.8067) is reported from a single training run and lacks multi-seed variance estimation. While the baseline configuration is validated across five seeds (0.8019 ± 0.0050), the optimal configuration identified through hyperparameter search has not been similarly characterised.

Interpretability model mismatch. The Integrated Gradients attribution and gene set enrichment analyses (Section 4.4) were performed on a six-network EMGNNImproved model, but not on the best heterophily-aware model (AUPR = 0.824). Re-running these analyses on the strongest model would provide a stronger biological validation, but is outside the final experimental scope.

Transductive target-network input. CPDB topology and node features are included as input while CPDB labels define the test set, following the original transductive EMGNN protocol. Test labels are withheld, but the high learned CPDB weight (0.210) may partly reflect access to target-network structure. A stricter leave-target-network-out evaluation is needed before interpreting the weights as intrinsic database quality.

Legacy split provenance. The experiments reported here predate the gene-level JSON split manifest introduced during the reproducibility audit. Although train and test membership originated from the benchmark masks, the historical validation subset was not saved independently. The current submission therefore reports these comparisons as explicitly labelled legacy evidence rather than as fixed-split revalidation results.

Zero-baseline IG attribution. Integrated Gradients uses a zero-vector baseline for feature attribution. Because the reported experiments use aligned raw benchmark features without additional Z-score standardisation, zero does not have one uniform biological interpretation across all modalities. Baseline sensitivity analysis was not performed.

Out-of-scope extensions. Four implemented techniques—pathway hypergraph integration, HIPGNN anomaly detection, PINNACLE pretrained embeddings, and GNNExplainer—await external dataset preprocessing or further debugging. They are retained as optional future work and are not used to support the conclusions of this paper.

Comparison scope. Direct numerical comparison with recent methods (SGCD, DISHyper, DGHNN, deepCDG) is limited by differences in evaluation protocols, dataset versions, and train/test splits. Standardised cross-method benchmarks on identical data splits would enable more rigorous comparison.

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Code and Data Availability

Source code, experiment scripts, compact result tables, and the manuscript are available at `https://github.com/EthyleneC2H4/NUS-Capstone`. Trained model checkpoints are not currently distributed through the repository. Multi-omics features and PPI network data are from the EMOGI benchmark (Zenodo: `https://doi.org/10.5281/zenodo.7654890`). MSigDB Hallmark 2020 gene sets were obtained from `https://www.gsea-msigdb.org`.

Author Contributions

Y.W. conceived the project extensions, implemented the software, conducted the experiments, analysed the results, and wrote the manuscript.

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Conflict of Interest

The author declares no competing interests.

Ethics Statement

This computational study used previously published, de-identified benchmark data and did not involve new human-participant recruitment or animal experiments.

AI Assistance Disclosure

Generative AI tools were used for code review, documentation support, and language editing. The author reviewed the generated material, verified the reported evidence, and retains responsibility for the implementation, analysis, and manuscript.

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